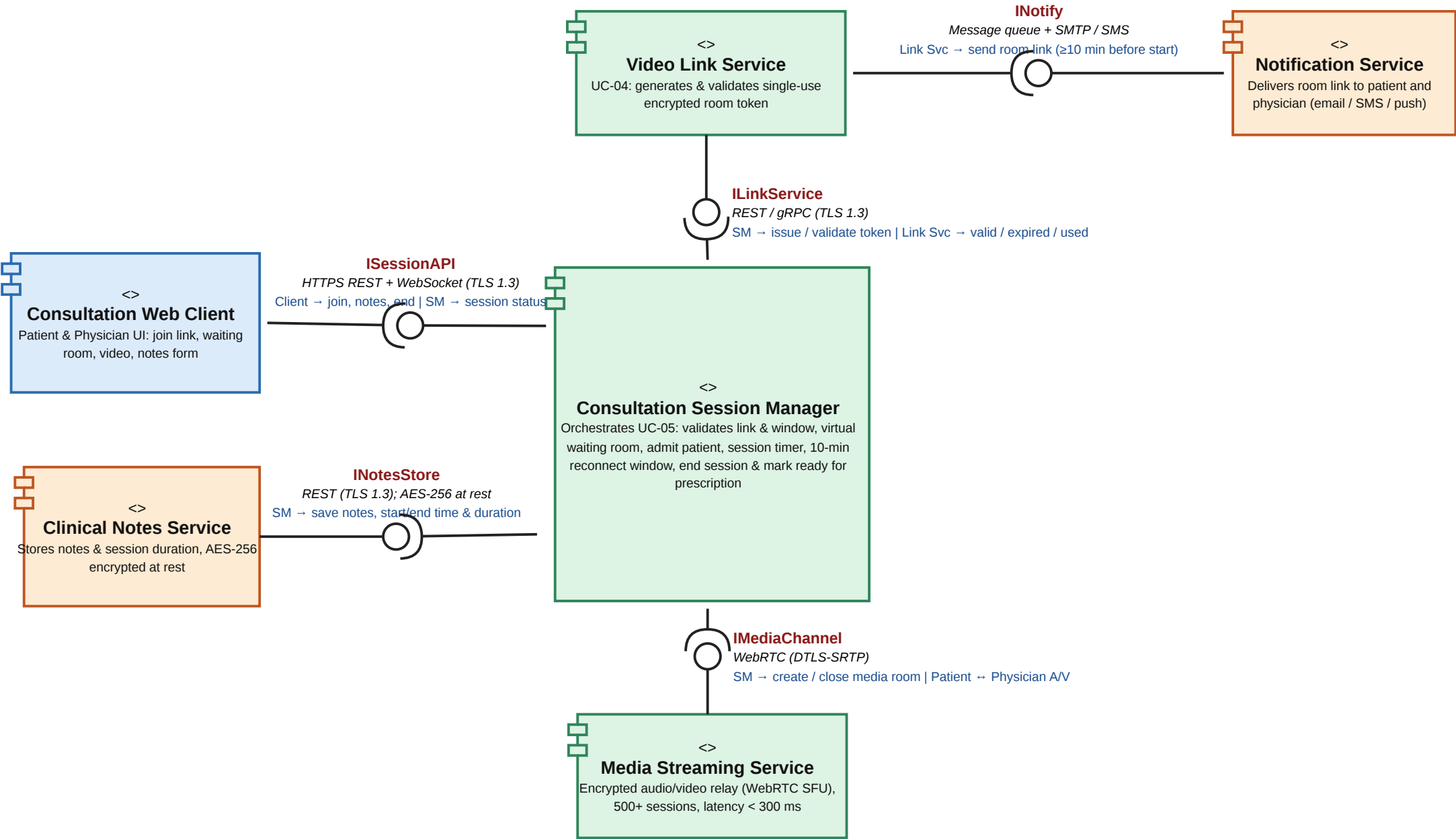


Lab 3 — UML Component Diagram: Tele-Consultation Session (Set 3, UC-04 & UC-05)

Telemedicine Slot Booking & Prescription Portal | Architecture: Microservices | Harshith R, PES1UG24AM117



Lab 3: Architecture Selection & Written Justification

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Architecture Selection

We chose **Microservices Architecture** for the **Tele-Consultation Session System (Set 3)**. The system is split into eight independently deployable components:

Consultation Web Client, Consultation Session Manager (UC-05), Video Link Service (UC-04), Media Streaming Service, Clinical Notes Service, Notification Service, and the neighbouring Booking Service (Sets 1/2) and Prescription Service (Set 4). They talk only through seven interfaces: ISessionAPI, IBookingStatus, ILinkService, INotify, IMediaChannel, INotesStore and ISessionCompletion.

Style Comparison

Style	Strength for this scenario	Weakness for this scenario	Verdict
Layered	Simple, clear separation of UI / logic / data	One shared stack: video load cannot be scaled alone; a layer failure stops everything	Not suitable
Client-Server	Central control, easy data consistency	Single server is a bottleneck and single point of failure for 500 live sessions	Not suitable
Microservices	Independent scaling, fault isolation, separate security boundaries	Network latency, operational complexity (acceptable trade-off)	Selected

Two Reasons for the Choice

1) Scalability for video (NFR-002) : The portal must support at least 500 concurrent video sessions under 300 ms latency. Video relay is by far the heaviest workload, while link generation and notes saving are light. Microservices let us scale the Media Streaming Service alone at peak clinic hours instead of replicating the whole application.

2) Fault isolation and availability (NFR-003, FR-004, UC-05 alternate flow 5a) :

A failure in the Notification Service or Prescription Service must not end a consultation already in progress. With separate services, the Session Manager keeps running, queues the link or the completion event, and still honours the 10-minute reconnection window. A layered or single-server design would let one fault take down every live session.

Security Advantage

Sensitive data is confined to small, separately protected services (NFR-001). Only the Clinical Notes Service holds clinical notes, encrypted with AES-256 at rest, and only the Video Link Service issues and validates single-use tokens (FR-004). All inter-service calls use TLS 1.3 and the video channel uses DTLS-SRTP, so a compromise of one component exposes far less patient health data than a shared database in a layered monolith. Each service also needs its own authentication, so there is a clear access-control point at every interface.

Performance Benefit

Media Streaming runs as its own service (a WebRTC relay), so audio/video traffic never competes with booking checks or notes storage for CPU and bandwidth.

It can also be placed close to users and scaled horizontally on demand, which is what keeps end-to-end latency under 300 ms at 500 concurrent sessions. Slower tasks such as sending the room link and handing over to prescription authoring run asynchronously through a queue / event broker, so they never delay the live session.

Trade-off acknowledged: microservices add network hops and operational complexity; we accept this because the availability, security and 500-session targets outweigh it.